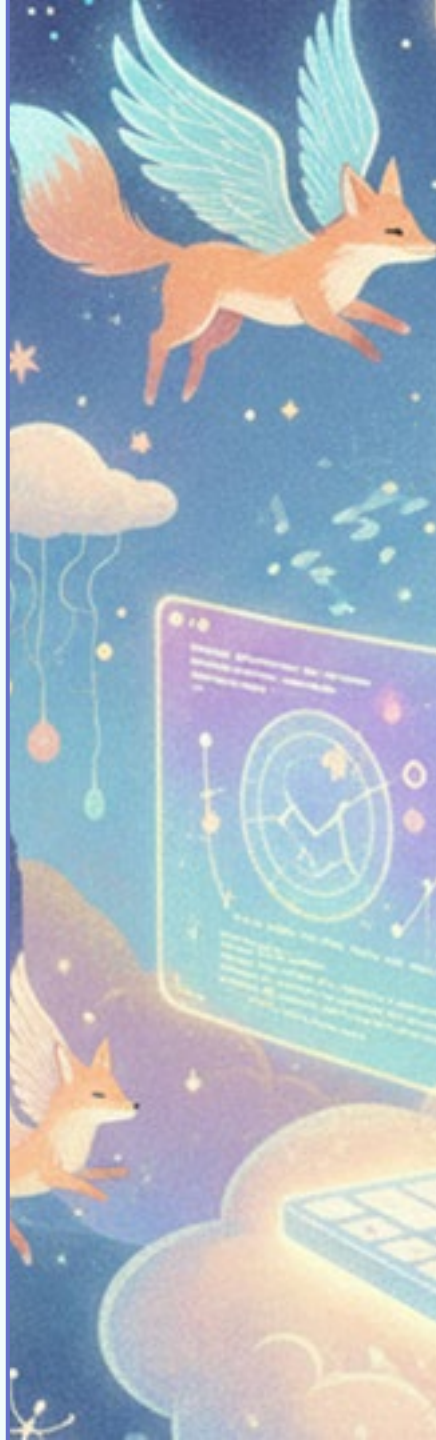
Welcome.

Everyone:

- Pull the updates from the course GitHub repo:
  - `cd <46120-PiWE repo>`
  - `git pull origin main`

NB:

- By attending this class, you consent to being recorded. Recording will be shared to this class and possibly other DTU students for training purposes.



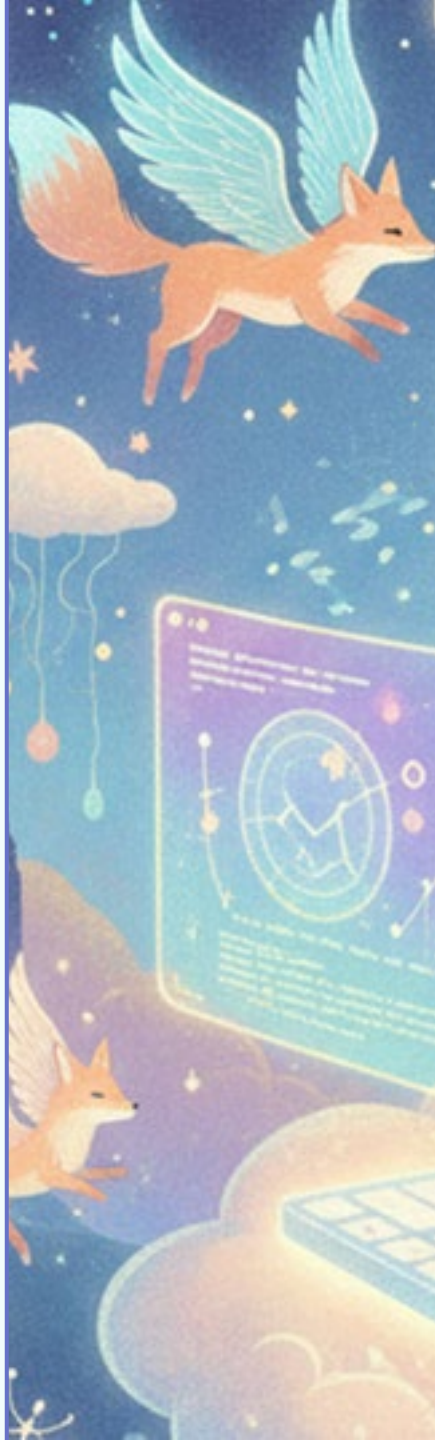
46120: Scientific Programming for Wind Energy

Environments and communicating code

Jenni Rinker

Round robin

Share solutions with your peers and give feedback.



20 minutes

Time to review and collaborate.

- 1 round of 20 minutes.
- IN PERSON TEAMS:
  - Form groups of 2-3. Try to talk with someone you haven't talked with yet!
- 5 minutes: chaos.
- 20 minutes: present/discuss homework. Today's feedback focus:
  1. How did they call solve_ivp? Same as you?
  2. How EFFICIENT is the code?
  3. How clean/understandable is the code?

- Afterwards: plenum discussion.
  - Be ready with questions!

PLENUM AT 09.20

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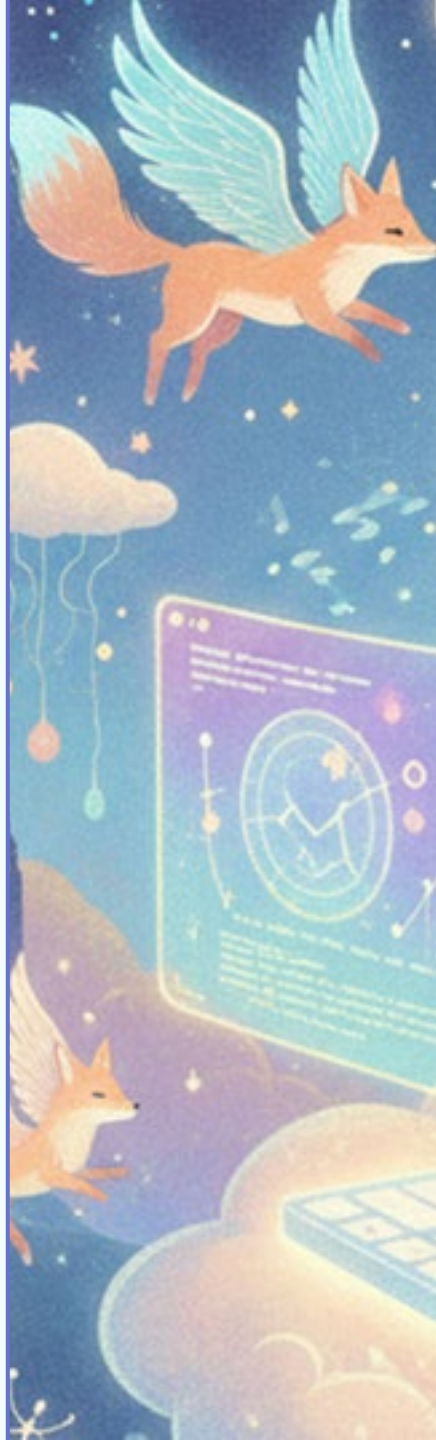
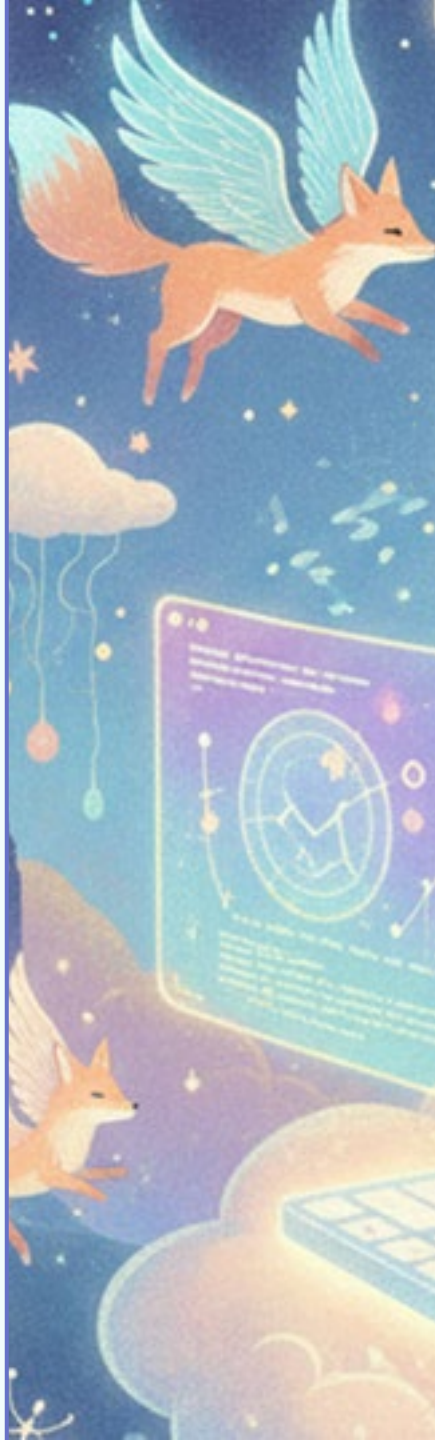


Notes in plenum.

- (notes here)



46120: Scientific Programming for Wind Energy

Environments and communicating code

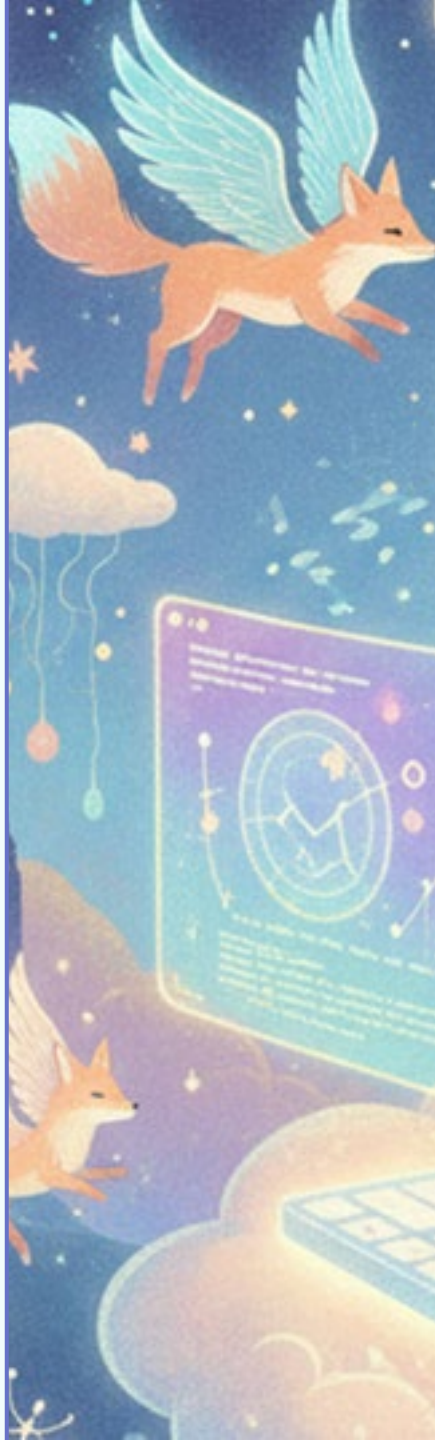
Jenni Rinker

Agenda for today.

- Pull new course material ✔

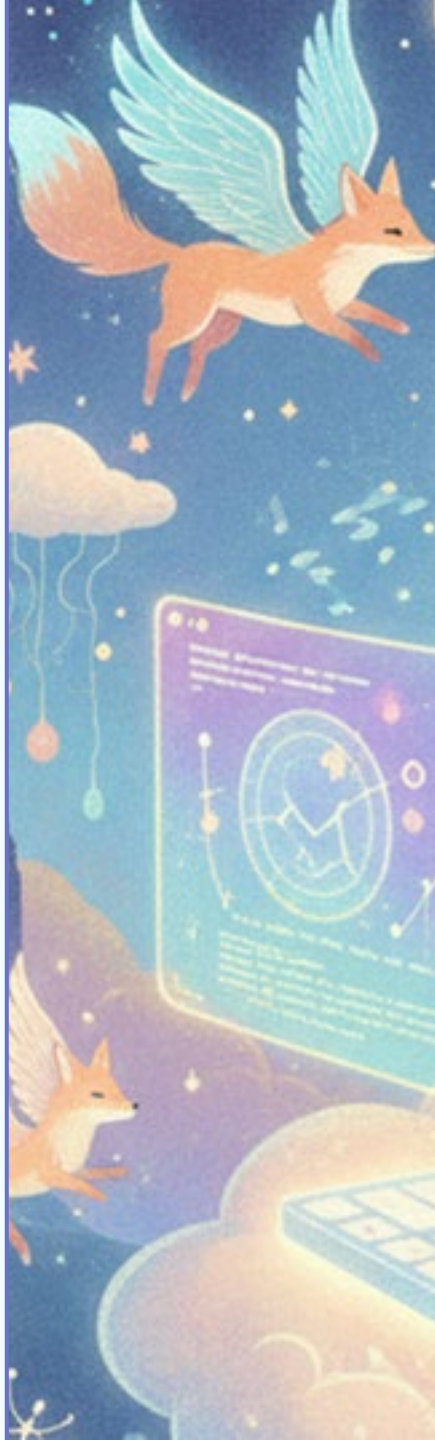
- Round robin. ✔
- Environments.
- Communicating code.

- Work on the CodeCamp project.
  - Due next Wednesday at midnight.



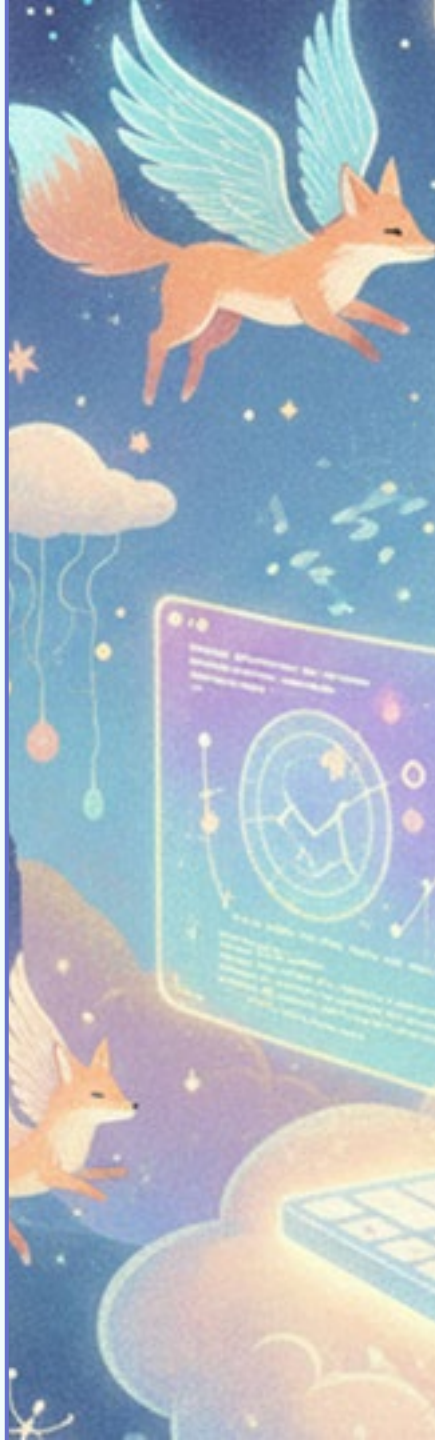
Environments

Keep stuff from messing up other stuff.



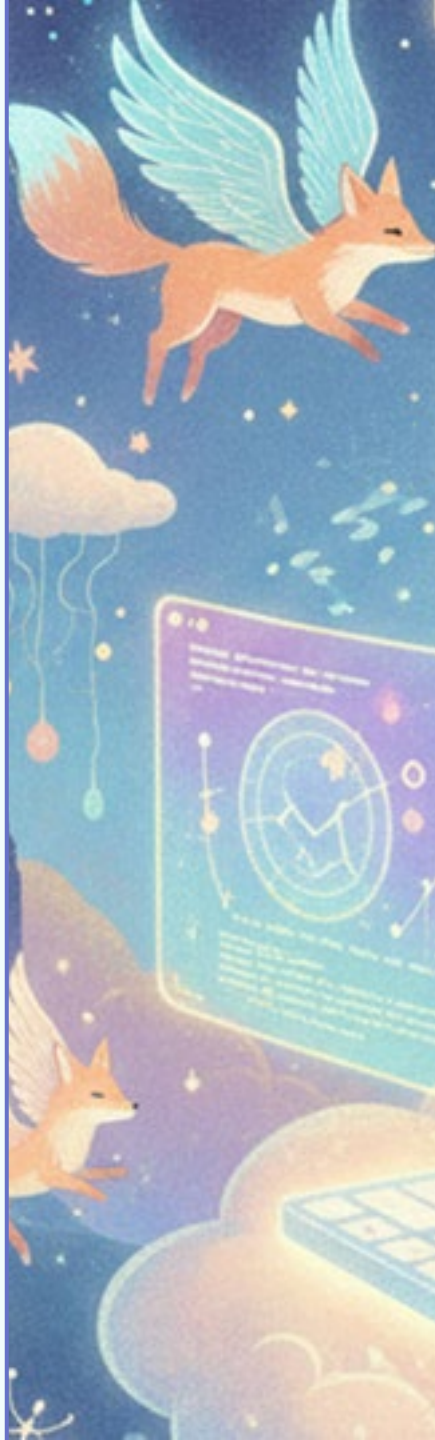
Problems with changing package/Python versions.

- You may see:
  - Tests that pass or fail depending on the Python version.
- Some other common situations:
  - Two packages require different versions of a required package.
  - A package requires a very specific order of required-package installation. You install another package, which messes everything up.
  - You have developed a package and need to test whether the dependencies get installed correctly.**
- Similar to branches in repos, we can use *environments*, which isolate specific versions of Python/packages to ensure code runs smoothly.



Environments are isolated spaces.

- Similar to branches in a repo.
- Allows you to create a special space, possibly with specific Python version, where you can install any packages you need.
- Two common ways to create environments:
  - conda -- allows different Python versions
  - Python virtual environments (venv) -- tied to a specific Python kernel
- You can switch between environments in the Anaconda Prompt and in VS Code.
- We'll learn conda here, but you'll see venv on the gbar cluster.
  - A comparison of conda versus venv is given in the appendices.



How to create/activate a PiWE environment.

Create/set up environment

1. Open an Anaconda Prompt and cd into your team repo.
2. Create a new environment with the name “piwe” and Python 3.11:

   ```
   conda create -n piwe python=3.11 -y
   ```

3. Activate the environment:

   conda activate piwe

4. Install the packages we need:

   ```
   pip install numpy matplotlib scipy pandas pytest
   ```

Activate environment

In Anaconda Prompt

1. Open an Anaconda Prompt.
2. Activate the environment:

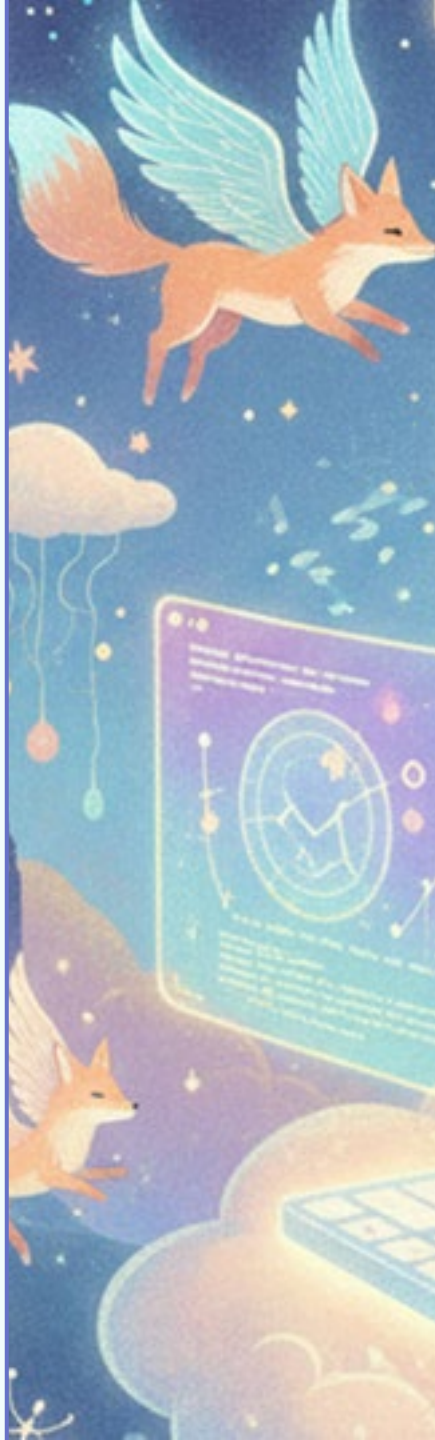
   ```
   conda activate piwe
   ```

3. Run pytest:

   ```
   pytest test_week4.py
   ```

In VS Code

1. Bottom bar, hold mouse over options until you find one that is a path to `python.exe`.
2. Click that, then select piwe from the drop-down menu.

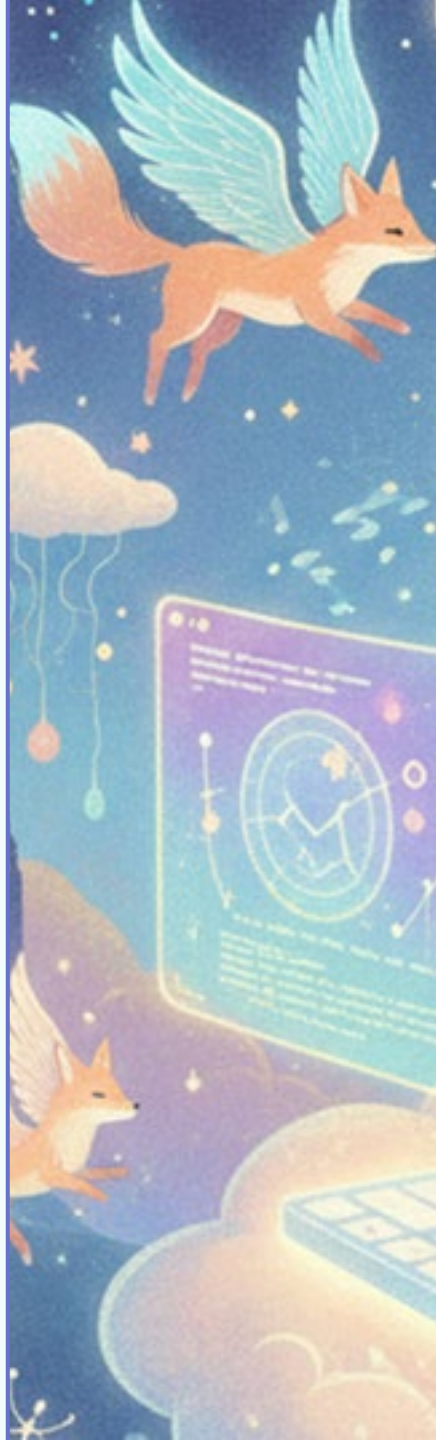


How to use environments in PiWE.

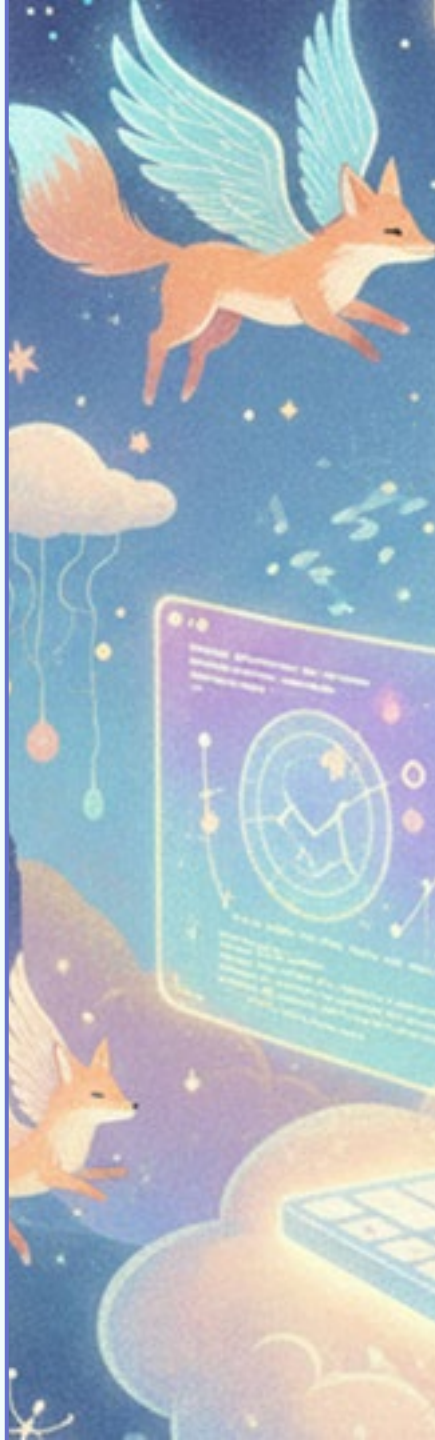
- For CodeCamp project, not required.
  - You are welcome to use it if you want your tests to pass.

- For final project, will be essential.
  - You will want to test installing your package in a clean environment, so you know that everything is installed correctly.

Questions?

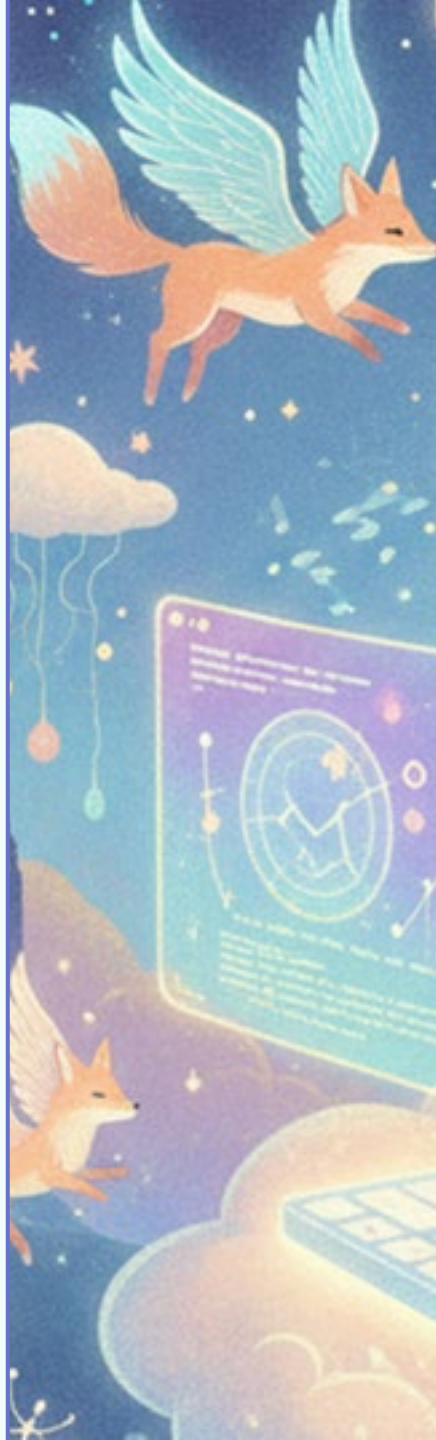


Communicating code



Part of programming is thinking programmatically.

- Three pillars of Computational Literacy [1]:
  - Cognitive CL: Breaking down a problem into steps.
  - Material CL: Turning those steps into code.
  - Social CL: Communicating/collaborating on code.
- Thinking programmatically is useful for both writing code and for communicating code.
- Three main tools we'll present:
  - Pseudocode
  - Comment prototyping
  - Code diagrams



Pseudocode.

- As the name suggests, **pseudocode** is fake code.
  - Useful for sketching out logic with pen and paper.

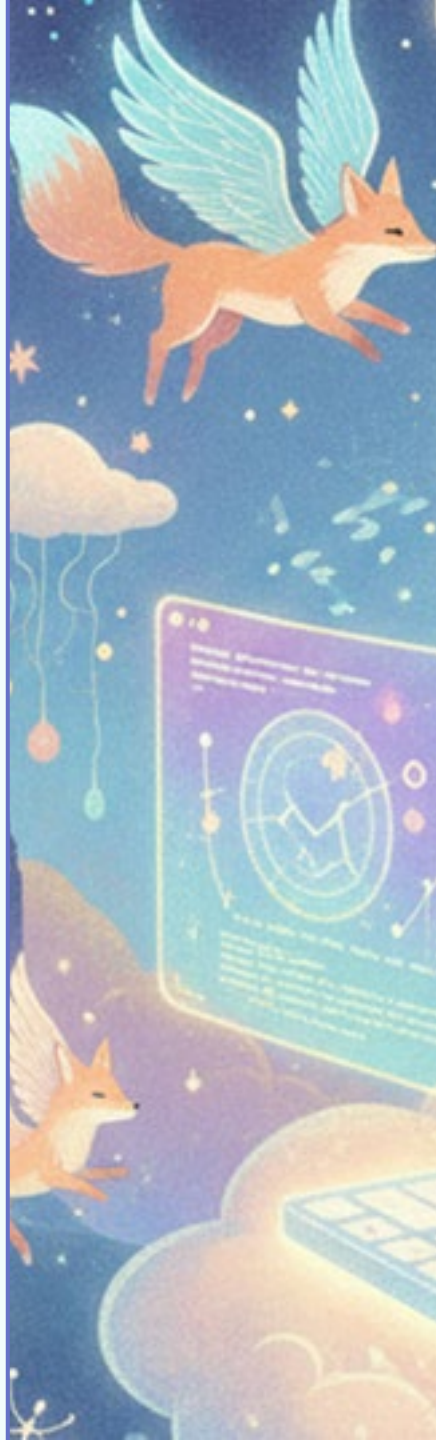
- No formal syntax, so you can develop your own style.

- A personal example

  combining 6

  10-minute files to 1

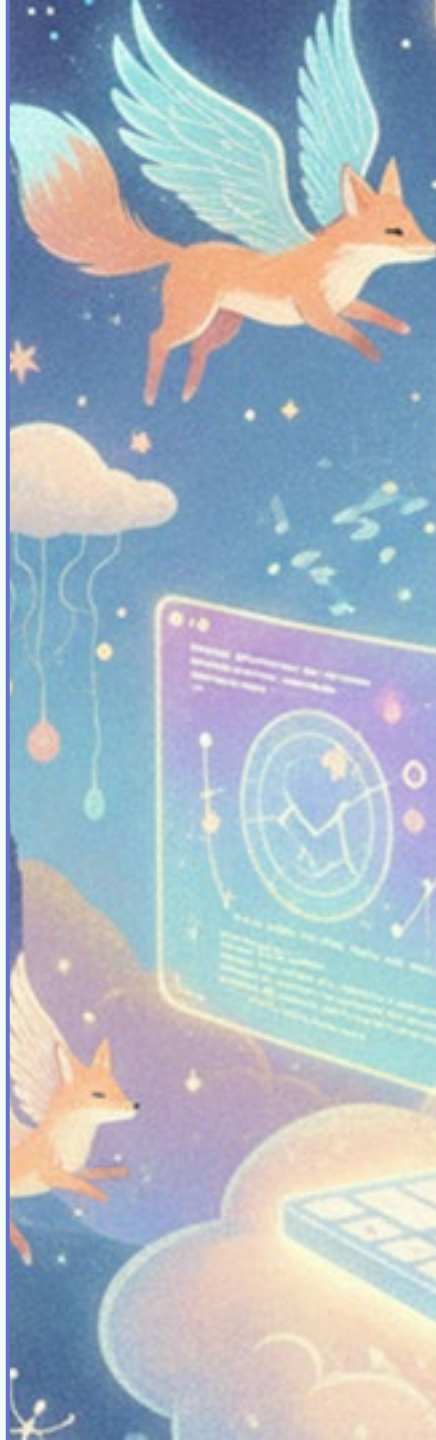
  1-hour dataframe.



Comment prototyping.

- This is a Jenni™ thing.
  - It's not on the internet.
- Helps if you're staring at a blank .py file and have programmer's block.
- Simple concept: prototype your code by writing the comments first.
  - Nice thing: comes right from pseudocode.

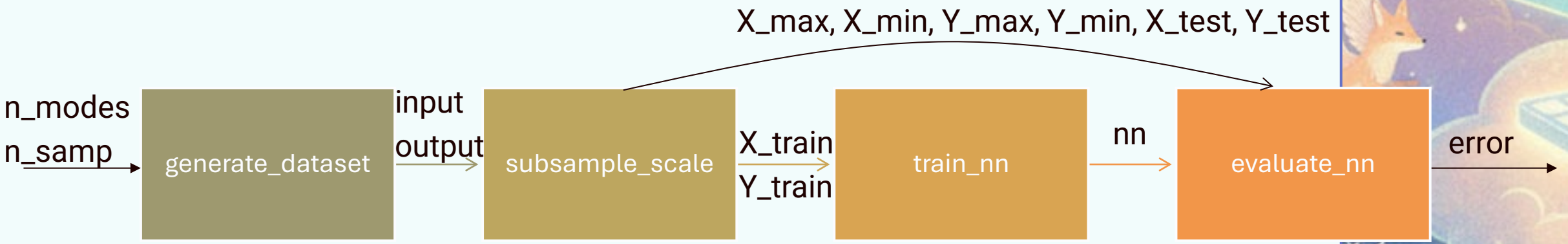
```
# define path to data directory

# get list of files in directory

# initialize empty dataframe

# loop over files

        # load the 10-minute data

        # update column names

        # assign to master df

# reset time index to start from 0
```

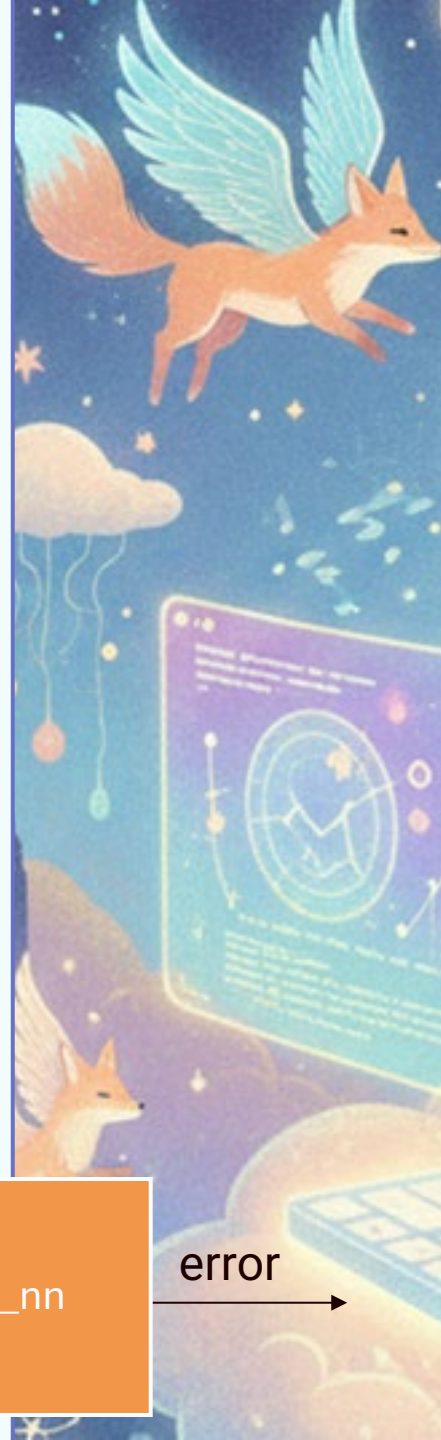


Code diagrams.

- For larger code, you may want to communicate/discuss/review the **architecture**.
  - I.e., how the code is structured, what are the inputs/outputs of your functions, etc.
- There isn't really a formalized way, so there is flexibility.
  - Although class diagrams [4] are good for classes.
- Could of course use your black-box diagrams.
  - Simple example with a neural-net training:

Potential to combine with pseudocode

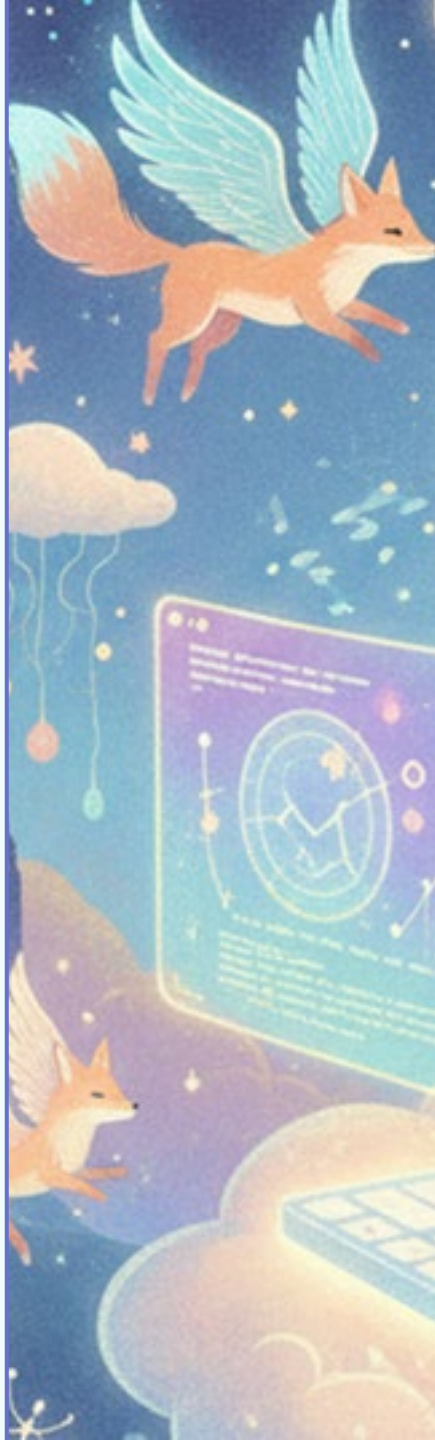




In short.

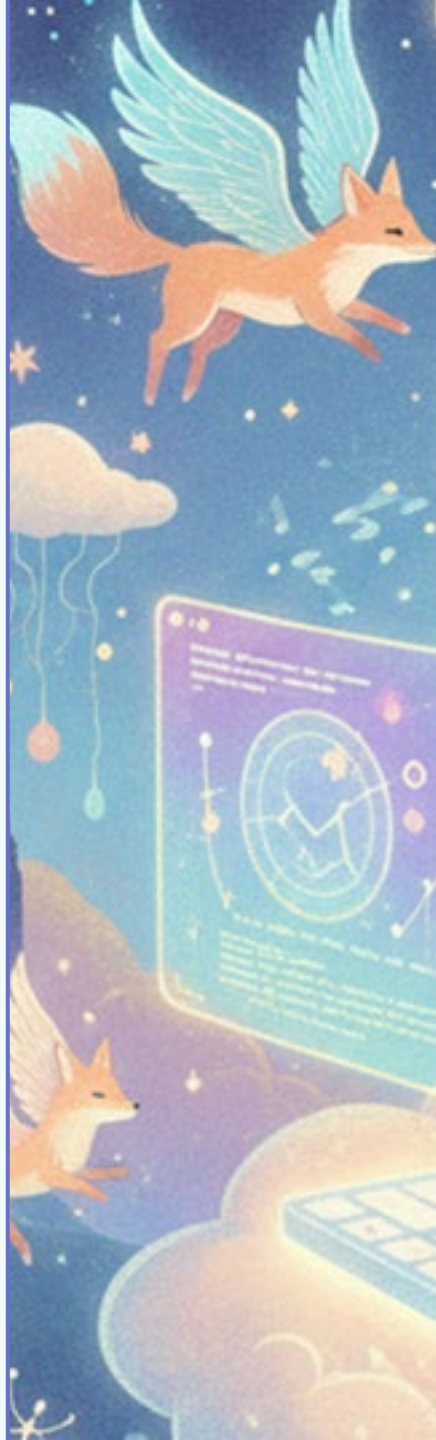
- There are many ways you can communicate about your code.
- For CodeCamp, your `README.md` file must have a quick-start guide and explanation of how the code works.
  - Target audience is a fellow student who has freshly cloned the repo. Assume that they have not taken this class but have the same Python/terminal skills as you.
- The sky is the limit!

Questions?



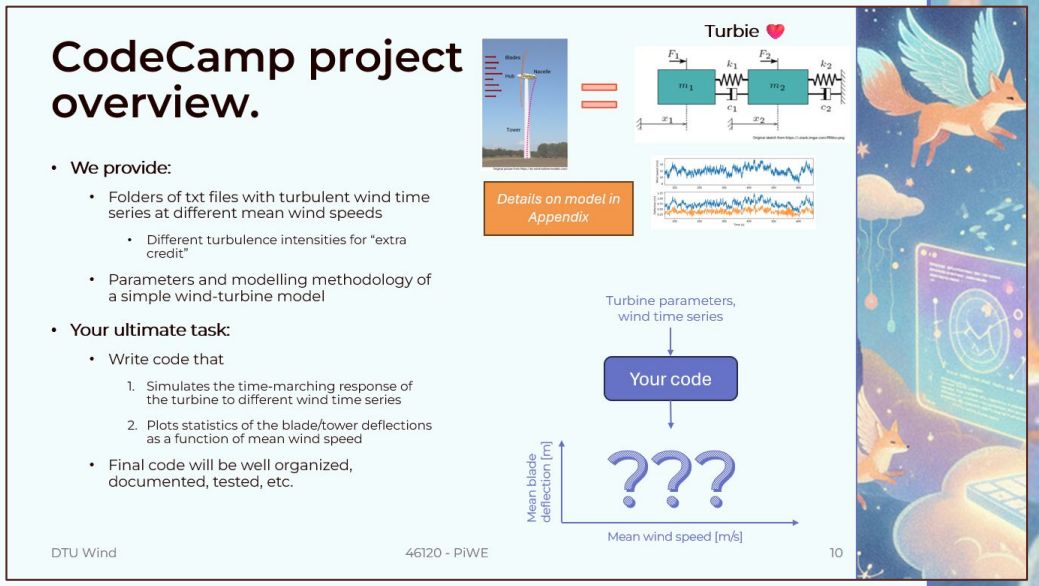
Homework for this week

Turbie here we coooooommmmmeeeeee!



Your task (Part 1).

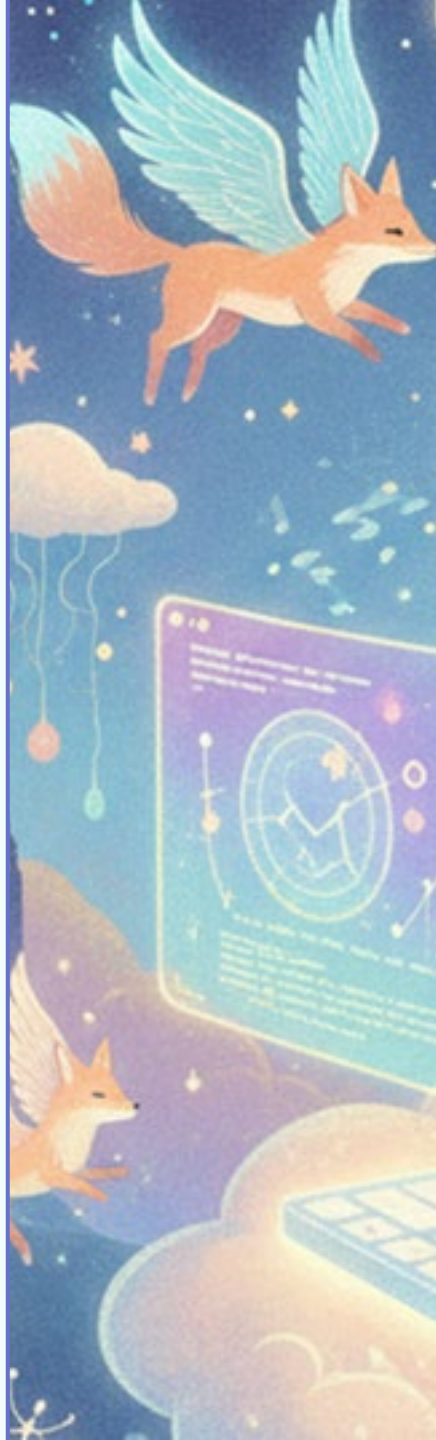
- "Finish the CodeCamp project." Repos lock 23:59 Monday March 9.
- Functions in Week 3 and 4 are the base of Turbie.
- Your team needs to design and implement the remaining code to create the requested plots.
  - Structure of the code, feature branches, etc., is up to you!
  - Your project "is done" when you satisfy requirements in Week 6.
  - Remember this slide from Week 3:



Your task (Part 2).

Instructions also in Week 6

- After CodeCamp is turned in but before Week 6 (Thursday March 12, 09:00).
- Your team collates and submits 3 (or 2) feedback forms.
  - NB: This is part of passing CodeCamp! Must also present feedback in Week 6.
  - Repo links for other groups are in CodeCamp sign-up on Learn.
- STRONG SUGGESTION.
  - Split evaluation amongst team members by rubric item.
  - I.e., Team Member A evaluates commit history for all 3 teams.
  - Use codecamp-eval-helper.xlsx to track/combine your feedback.
- When xlsx filled out, submit evaluation form using link on Learn.
  - !!IMPORTANT!!!
    1. Check “send me an email receipt of my responses” before submission
    2. Click “Save my response to edit” after submission in case you made a mistake.



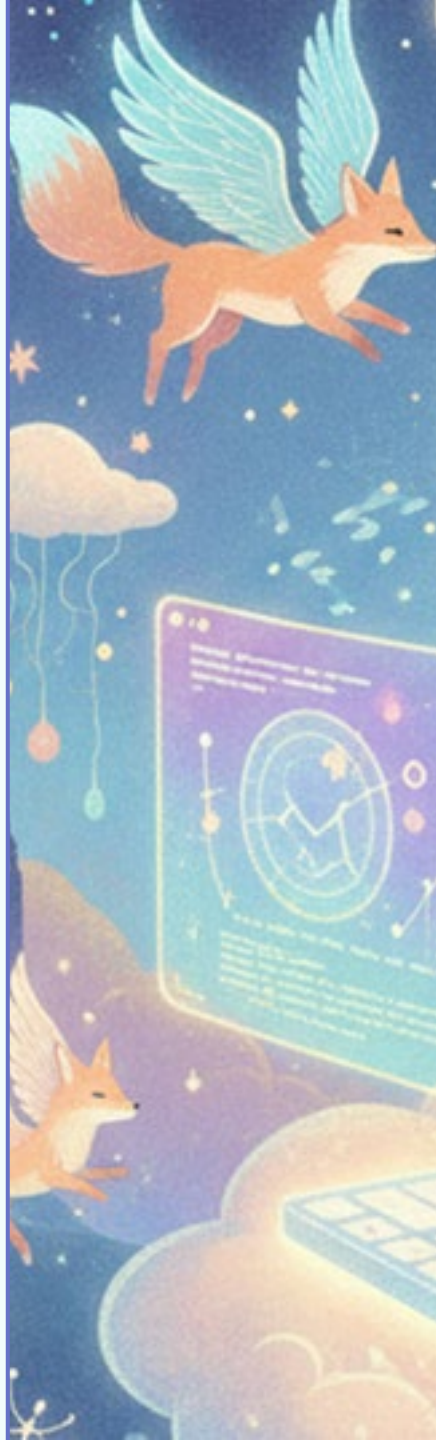
*Remember, you're expected to work **about 6 hours outside of class**. Schedule accordingly.*

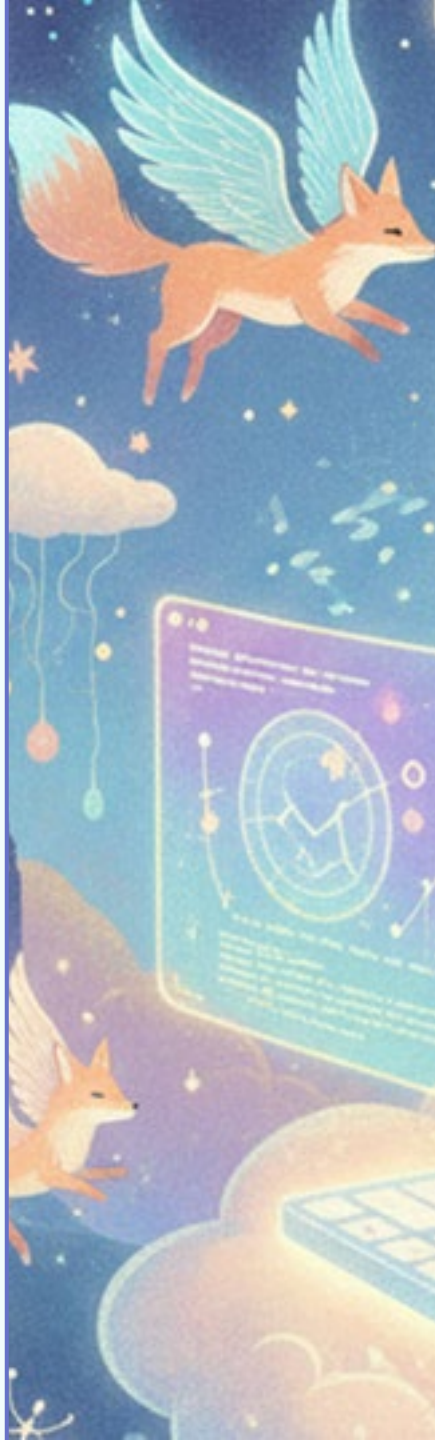
Homework.

- Description/requirements in week06 subfolder on GitHub.
- We'll open BORs in a minute.
- **To get help during class**: Post in Slack / #debugging if you want a TA to enter your BOR or come find your group.

Any questions?

Then see you next week when you present your feedback. ☺



References.

1. DiSessa, Andrea A. *Changing minds: Computers, learning, and literacy*. Mit Press, 2000.